CATALYST 2026 · TRANS ONTARIO · SMARTER FREIGHT, SMARTER ROUTES

The right truck for every new pickup, in seconds

A live dispatch decision layer between Elite (orders) and Samsara (trucks), built and tested on Trans Ontario's own data.

Live app https://trans-ontario-dispatch.onrender.com/dispatch

Code https://github.com/kavuri-sai-vardhan-simha/trans-ontario-dispatch

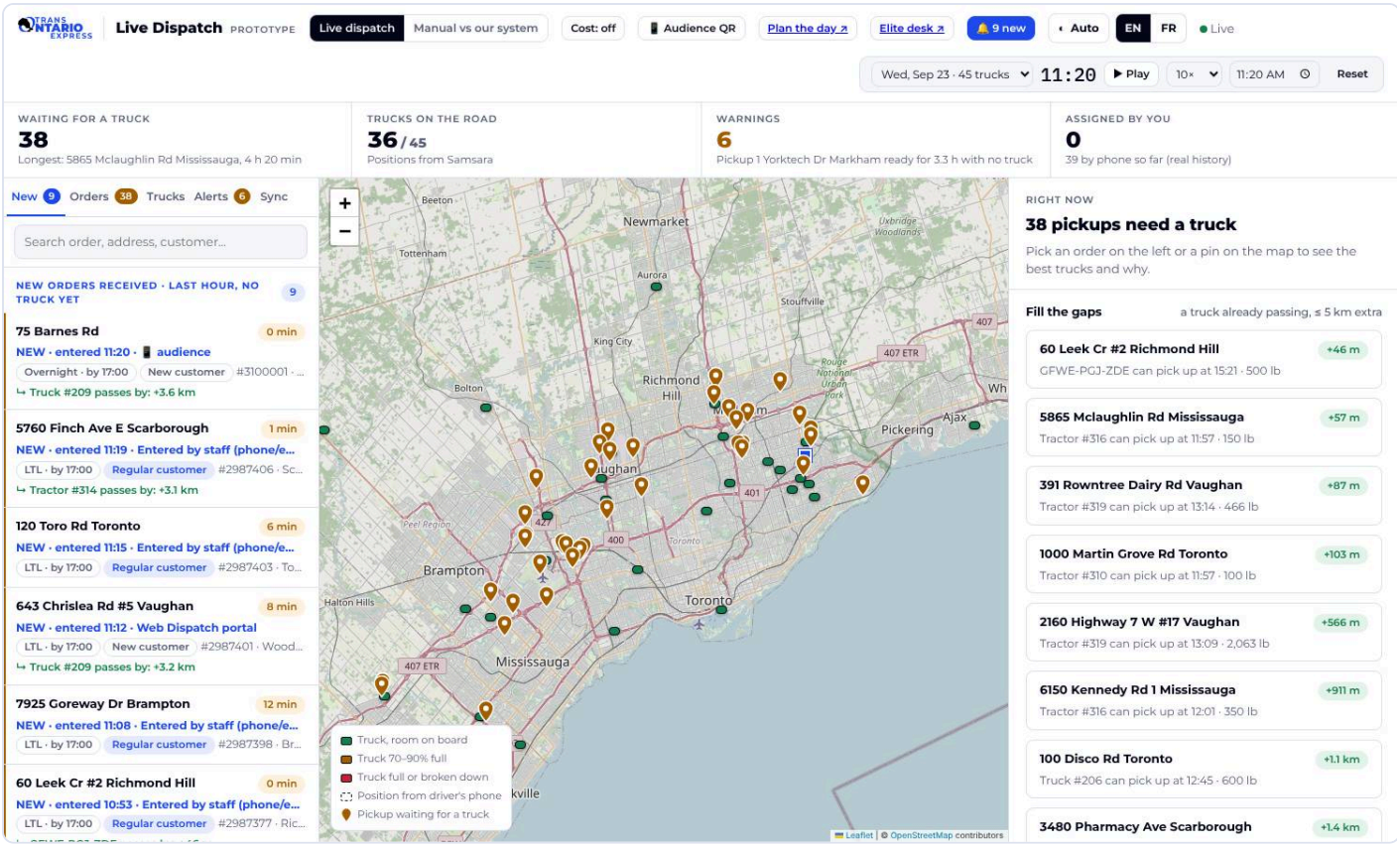
The live app runs on a free host: the first visit after a quiet period can take about a minute to wake up.

1. The problem

Trans Ontario hands out about **160 pickups a day by phone**. The tracking screen isn't live, so dispatchers phone driver after driver: where are you, do you have room, can you make it by the cutoff? **81%** of pickups are booked the same day and **82%** are assigned between 9 am and 3 pm, while trucks are already out. In one real case (Sep 29, 111 Gibraltar Rd, Vaughan) a truck with room was delivering **110 m away**; the pickup went to a truck 137 km away and the freight waited **5 h 44 min**.

2. What we built: five connected screens

Screen	Who uses it	What it does	Open it
Dispatcher	Dispatch office	Live map of every truck; **New** orders with a count; for any order, the best trucks with reasons and the trucks ruled out; tap a truck to see it on the map; assign in one click; warnings, driver concerns and disruptions; *Manual vs our system* comparison.	/dispatch
Elite order desk (simulator)	Office staff	Stand-in for Elite: the day's real orders appear when they were entered; staff can enter a phone/email order; each order shows its truck. In production our layer reads Elite directly.	/elite
Driver app (inside Samsara Driver)	Driver, on a phone	The new pickup is already on the route: **OK, got it** or raise a concern. Report free floor spots, problems (sick, truck, traffic, freight not ready, failed delivery), share phone GPS.	/driver
Plan the day	Dispatcher	Plans every stop of the day on every truck; **Model's plan vs Actual dispatch** on the map; issues, disruptions, rules; live orders assigned by the dispatcher appear on their truck automatically with every rule ticked; truck search.	/plan
Booking page (QR)	Customer / audience	Stands in for Trans Ontario's Web Dispatch portal: pin a pickup and delivery, book, then track it as the dispatcher assigns a truck.	/book



Dispatcher: live map, new orders, warnings. Replay of Wed Sep 23, 2026 (45 trucks).

3. How the model chooses a truck

For each truck, the system tries every place the pickup could fit in the truck's remaining route and asks three questions. Only trucks with **three YES answers** are shown.

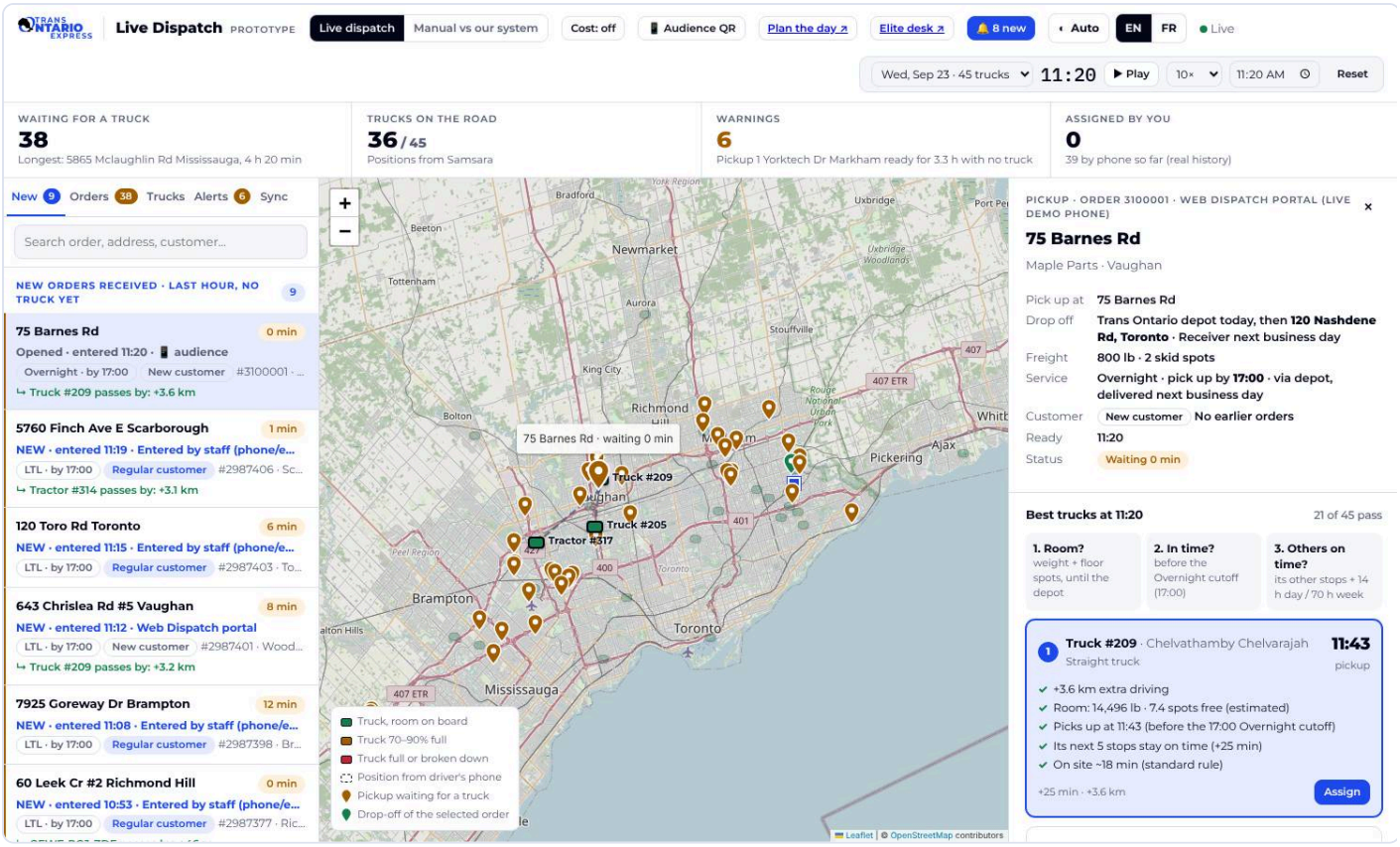
1 · Is there room?
Weight and skid spots at the **fullest point** between the pickup and where that freight comes off the truck, so a truck that fills up later in its route is never over-promised.

2 · Can it get there in time?
Before the service cutoff (Rush 3:30 pm, Out of town 1:30 pm, others 5 pm), within shipper hours and appointments; same-day freight still delivered today.

3 · Is everyone else still on time?
No other stop on that truck may become late to fit one more pickup.

Also always checked: driver hours (14 h on duty, 70 h in 7 days), truck type, tailgate for no-dock sites, height, dangerous goods (TDG) never stacked. A rule is never broken.

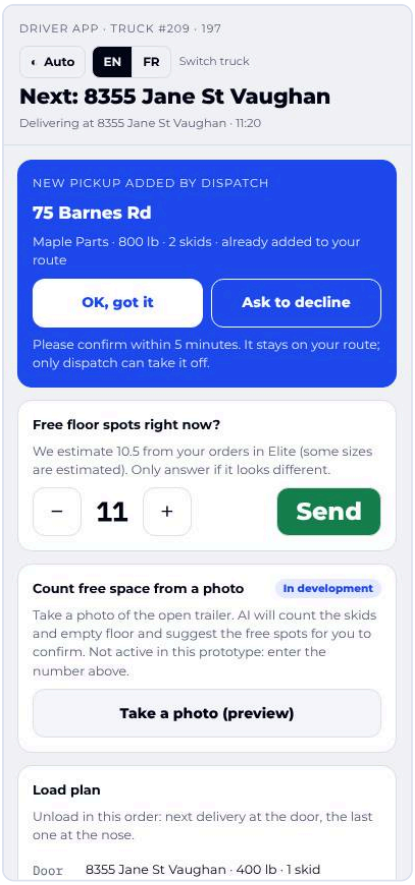
Ranking of the trucks that pass: customer wait (regular customers, found automatically from past orders, count 1.5×) + extra driver minutes + extra km. Each suggestion shows its reasons (room left, pickup time, "its next 4 stops stay on time", time on site) and how many trucks could take it.



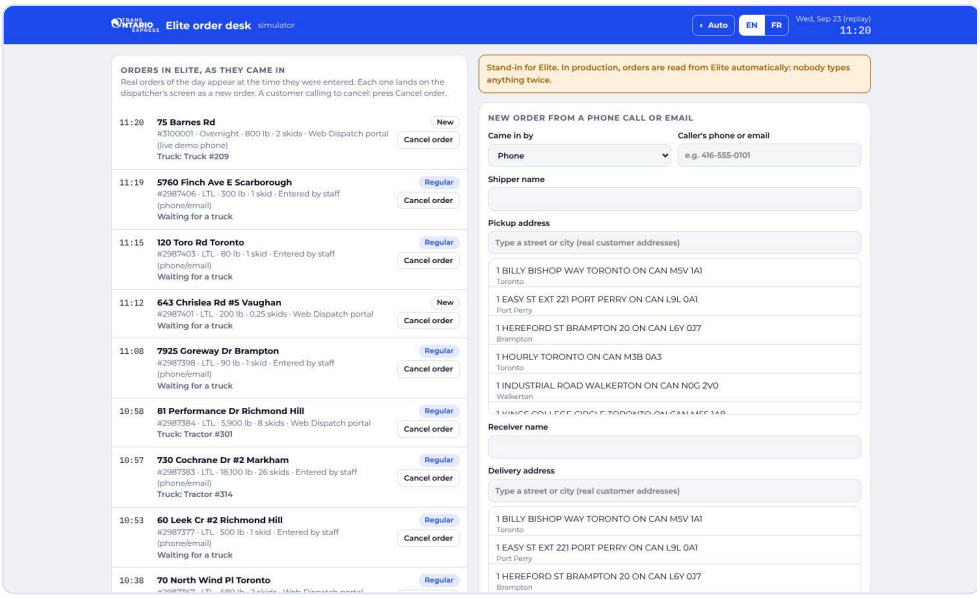
An order opened: pickup and drop-off pins, size, service, customer type, the three checks and the best trucks with reasons.

4. After the dispatcher assigns

- The pickup goes onto the driver's route in Samsara; Elite, the dispatcher, the driver app, the day's plan and the customer's tracking update together.
- The driver taps **OK, got it** or raises a concern (no room, too late, hours, other). **No OK in 5 min:** automatic reminder. **10 min:** the dispatcher is told to call.
- The dispatcher decides on a concern: approve (the order goes back to the list for another truck) or keep it. Reassign or cancel, and every screen follows.



Driver app: new pickup on the route, OK or ask to decline, free spots.



Elite desk: orders as they arrive, each with its truck.

5. When things go wrong

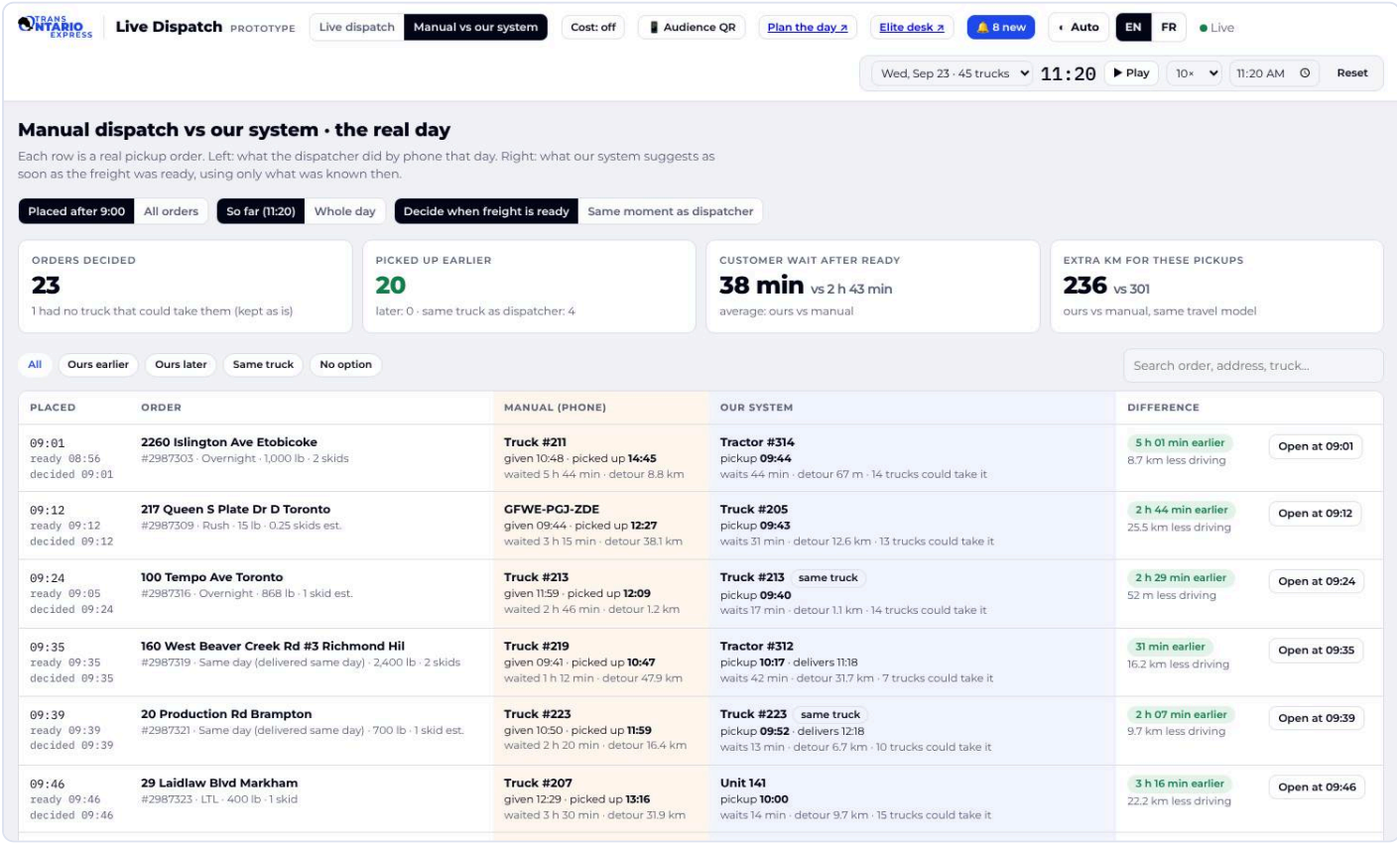
Disruption	What the system offers
Truck breaks down	Which truck can meet it and take the load; which pickups move to other trucks; before/after shown.
Driver calls in sick	A spare driver with the right licence on the same truck, or share the work with other trucks.
Traffic delay	New ETAs, which promises would break, which pickups another truck could take.
Freight not ready	Wait for it (and its effect) or leave and give it a truck when it's ready.
Failed delivery	Bring it back for next business day, or re-attempt today if it still fits.
Customer cancels	Removed everywhere; the truck's room is freed.

6. Results on 10 real days (Sep 21 – Oct 2, 2026)

Measure	Our system	Real dispatch
Time to check every truck for one order	**3 ms**	phone calls
500 orders in one day (stress test)	**1.5 s** in total	—
New orders, live: picked up earlier / later (by >15 min)	**599 / 202** of 945	—

New orders: average customer wait after the freight is ready	**37 min**	1 h 35
New orders: km for those pickups	15,657 (+4%)	14,987
Whole-day plan: trucks sent out	**277** (−35%)	428
Whole-day plan: late stops	**159** (−59%)	387
Whole-day plan: km	82,761 (same)	83,674

How to read it honestly. The live test replays the 1,072 pickups given out during the day one at a time, using only what was known then; even deciding at the exact moment the real dispatcher did, ours picks up 466 earlier vs 202 later. The whole-day plan sees all orders at once, so it is a best case. Both sides use the same travel model (calibrated on Trans Ontario's GPS) and the same rules; Trans Ontario vehicles only, no partner carriers.



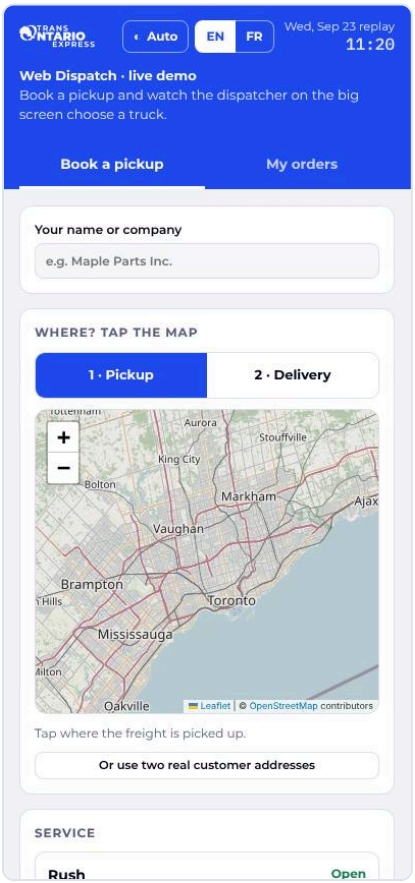
Manual vs our system: each real order, what the dispatcher did by phone vs our suggestion, with the difference.

7. What sets it apart

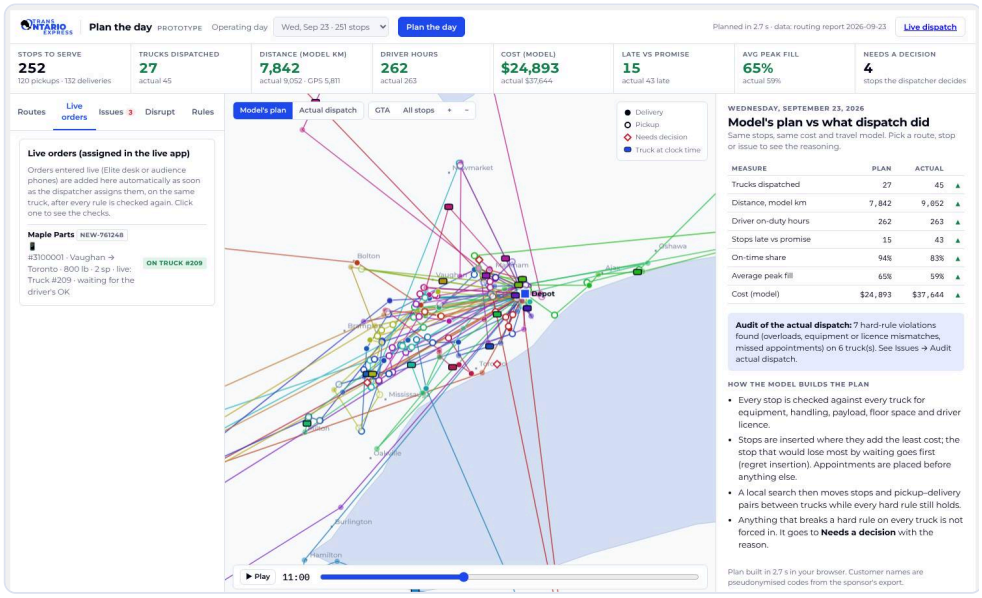
- **Built for same-day work.** Most routing tools plan in the morning; here 81% of pickups arrive during the day. We handle each new order live, and keep the day's plan in sync.
- **Explains every choice.** Reasons for the best trucks and for every truck ruled out, so dispatchers can trust it and still decide.
- **Never breaks a rule** to look good: room at every stop ahead, cutoffs, appointments, driver hours, equipment.
- **No new apps.** Orders stay in Elite, drivers keep Samsara; the layer sits in between.
- **Proven on their own data,** with a side-by-side comparison against what really happened, order by order.
- **Customer-aware:** regular customers are recognised automatically and wait less.
- **AI where it helps:** time on site learned from 1,478 real stops (tested on days it never saw: error 18 vs 19 min, bias 5 vs 12 min).

8. How to view it (no install)

1. Open https://trans-ontario-dispatch.onrender.com/dispatch (allow up to a minute on the first visit). The replay shows Wed Sep 23 at 11:20.
2. Click 📱 **Audience QR** and scan it with a phone, or open /book; book a pickup. Or enter one on the Elite desk.
3. In the dispatcher's **New** tab, click the order: see the three checks and the best trucks. Tap a truck card to see it on the map; click **Assign**.
4. Open /driver, pick that truck: the stop is on the route; tap **OK, got it**.
5. Open **Plan the day ↗**: the stop is on that truck's route with every rule ticked (tab *Live orders*). Flip **Model's plan / Actual dispatch**.
6. Open **Manual vs our system** for the comparison on real orders. **Reset** returns to the start of the day; the day picker offers all 10 real days.



Booking page (Web Dispatch stand-in).



Plan the day: model's plan vs actual, live orders on their truck.

9. How to run it on your own computer

Needs	Python 3.10+ and internet (for map tiles). No Node.js needed: the screens are prebuilt.
Windows	Download the repo → `live-dispatch\run_prototype.bat` (installs FastAPI/uvicorn and opens the dispatcher).
Mac / Linux	`cd live-dispatch/backend && pip install -r requirements.txt && python -m uvicorn app.main:app --port 8000`

Addresses	localhost:8000/dispatch · /elite · /driver · /plan · /book
Public QR link (optional)	Put `cloudflared.exe` in `live-dispatch`; the app opens a free Cloudflare link for the QR. The public link opens the booking page only.
Host it	Render: New → Blueprint → the repo (`render.yaml` included).
Plan the day alone	Open `plan-the-day/index.html` in Chrome; no server.

10. Data, assumptions and limits

- **Data:** Trans Ontario's anonymised routing report (Samsara stops/GPS + Elite orders), Sep 20 – Oct 3, 2026: 10 operating days, 1,635 pickups, 2,530 stops. Truck sizes and cutoffs from trans-ontario.com (straight truck 12 skids / 15,000 lb; 53 ft trailer 26 skids / 44,000 lb).
- **Replay, not yet live:** the Sync tab shows exactly what would be sent to Elite and Samsara; connecting needs their access.
- **Estimated:** travel times (calibrated model, no live traffic; model km read higher than GPS, same model for both sides), truck heights, skid spots where Elite has none (recorded on only 35% of stops).
- **Flagged, not assumed:** drivers' dangerous-goods certificates are not in the data.

11. Built with

Python · FastAPI · SQLite · WebSocket (live updates) · React + Vite · Leaflet / OpenStreetMap · a JavaScript planner (regret insertion + local search) for Plan the day · stop-time model trained on the report. Repo: `live-dispatch/` (backend, web, data, scripts, docs) and `plan-the-day/`.

12. What comes next

Connect to Elite and Samsara · real road distances and live traffic · AI helpers: booking email → order form, travel times by hour, late-risk alerts, free space from a trailer photo, predicting tomorrow's pickups · customer email updates.